MIL-I-12123D

16 June 1967

SUPERSEDING
MIL-I-12123C
30 December 1963

MILITARY SPECIFICATION

INSECT REPELLENT, CLOTHING APPLICATION

(FORMULA M-1960)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers one type and grade of insect repellent for clothing application (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636	-	Box, Fiberboard.
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 141	-	Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

FSC 6840

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation D-86-62 - Method of Test for Distillation of Petroleum Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

PHARMACOPEIA OF THE UNITED STATES

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pennsylvania.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Composition.- The composition of the clothing application repellent shall conform to the ingredients in the weight proportions specified in table I (see 4.1.2).

TABLE I.- Composition of clothing application repellent

Ingredient	Percent by weight
2-Butyl-2-ethyl-1, 3-propanediol	30 ± 0.5
Benzyl benzoate	30 ± 0.5
N-Butyl acetanilide	30 ± 0.5
Polyoxyethylene (20) sorbitan monooleate	10 ± 0.1

3.1.1 2-Butyl-2-ethyl-1, 3-propanediol.- The 2-butyl-2-ethyl-1, 3-propanediol shall be a white crystalline solid, and shall conform to the physical and chemical requirements specified in table II when tested as specified in 4.2.2.

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TABLE II.- Chemical and physical requirements

Property	Minimum	Maximum
Color scale (Hazen)	---	250
Freezing point (°C.)	40	41.8
Free acidity as propionic acid, percent	---	0.05

3.1.2 Benzyl benzoate.- The benzyl benzoate shall be a clear liquid, and shall conform to the physical and chemical requirements specified in table III when tested as specified in 4.2.2.

TABLE III.- Chemical and physical requirements

Property	Minimum	Maximum
Color scale (Hazen)	---	250
Setting point (°C.)	18.0	
Specific gravity 20°/4°C.	1.100	1.120
Refractive index n/D 20°C.	1.560	1.570
Total organic chlorine, percent	---	0.15
Side chain chlorine, percent	---	0.05
Ester content, percent	98	
Free acidity as benzoic acid, percent	---	0.1

3.1.3 N-Butyl acetanilide.- The N-Butyl acetanilide shall be a clear liquid, and shall conform to the physical and chemical requirements specified in table IV when tested as specified in 4.2.2.

TABLE IV.- Chemical and physical requirements

Property	Minimum	Maximum
Color scale (Hazen)	---	250
Setting point (°C.)	22.0	

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TABLE IV.- Chemical and physical requirements (cont'd)

Property	Minimum	Maximum
Boiling range, percent:		
Up to 278°C.	---	5
Up to 283°C.	95	
Free acidity as acetic acid, percent	---	0.5

3.1.4 Emulsifier.- The emulsifier, polyoxyethylene (20) sorbitan monooleate shall conform to all the requirements of Polysorbate 80 as specified in the Pharmacopeia of the United States. In addition, this emulsifier shall contain not more than 0.2 percent water when tested as specified in 4.2.2.

3.2 Homogeneity.- The finished product shall show no separation of solid or liquid phase when tested as specified in 4.3.2.

3.3 Emulsibility.- The finished product shall form a creamy emulsion and shall readily re-emulsify when tested as specified in 4.3.2.

3.4 Labeling data.- The labeling data shall be as follows:

Insect repellent, clothing application formula M-1960

Active ingredients:	Percent
2-Butyl-2-ethyl-1, 3-propanediol	30
Benzyl benzoate	30
N-Butyl acetanilide	30
Inactive ingredient:	
Polyoxyethylene (20) sorbitan monooleate	10
	100.0

DIRECTIONS FOR USE

To impregnate clothing for protection against mites, fleas, ticks, mosquitoes and land leaches.

For hand impregnation. Caution: The tropical cotton uniform absorbs different amounts of repellent than the standard cotton or woolen uniforms.

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For this reason, it is necessary to vary both the concentration and the amount of the stock solution applied to the tropical uniforms. The following table gives the directions for mixing stock solutions and the amount to be used for each type of uniform. It is essential that these directions be followed to insure adequate protection and prevent over treatment.

PREPARATION AND USE OF STOCK SOLUTION

Woolen or standard cotton uniforms. Add one part M-1960 concentrate (by volume) with eleven parts water (by volume) and mix thoroughly. Apply mixture at the rate of 3.5 pints of stock solution per uniform (shirt or jacket, trousers, and socks).

Tropical cotton uniforms. Add one part M-1960 concentrate (by volume) with four parts water (by volume) and mix thoroughly. Apply mixture at the rate of 2.5 pints per uniform (jacket, trousers, and socks).

After preparation, the stock emulsion should be stirred occasionally to maintain the emulsion. Place one or more uniforms in a container and thoroughly wet it with the required quantity of mixed stock solution. Remove clothing from container and wring out excess emulsion. Do not re-use emulsion not absorbed by the uniform. Place uniform on a flat surface to dry. Do not wear uniforms until completely dry. Subsequent re-impregnations should be carried out as described above.

For machine impregnation. Clothing can be impregnated with M-1960 emulsions after washing or thoroughly wetting in fixed laundries or mobile laundry units. The procedure for impregnation and any subsequent re-impregnations is as follows:

Standard cotton uniforms. Add 1-1/2 gallons of concentrate to 4 gallons of water heated to approximately 100°F. and mix thoroughly. This quantity of emulsion will impregnate 60 pounds dry weight of cotton uniforms (20 fatigue or combat jackets and 20 fatigue or combat trousers. After the last rinse or after clean clothing is thoroughly wet, drain the washer and roll for 2 minutes to remove excess water from the clothing and to insure complete drainage of the reservoir. Add the prepared emulsion and roll for 5 minutes. Drain washer and remove clothing for drying. Extract for 3 minutes and tumble dry for 10 minutes at 200°F.

Wool uniform. Add 3 quarts of concentrate to 4 gallons of water heated to approximately 100°F. to impregnate 60 pounds dry weight of wool uniforms (14 shirts and 14 trousers). Impregnate as described for cotton clothing with the exception that drying should be carried out at not over 140°F.

Tropical cotton uniform. The process for treating tropical uniforms is the same as that for standard cotton uniforms except that five gallons of M-1960 concentrate with no water is used for 22 uniforms.

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Precaution. Untreated shorts should be worn under treated uniforms to prevent any possible skin irritation in the groin. Caps, when treated, should be held by the visor while dipping only the crown to prevent damage to the laminated visor by the emulsion.

Note: This material should not be relied upon to protect against mosquitoes on exposed skin.

3.5 Average net content.- The average net content per unit container shall be not less than specified.

3.6 Workmanship.- The finished product shall be a clean, clear, homogeneous solution free from all insoluble matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.1.2 In-process examination.- In-process examination of the batch mix shall be made to determine that composition formulation complies with table I.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except when otherwise indicated hereinafter.

4.2.1 Inspection of materials and components.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase documents.

4.2.2 Testing of materials and components.- The materials and components shall be tested for the characteristics specified in table V. The lot size shall be expressed in gallons. The sample size shall be a quart composite. The composite shall be made by thoroughly mixing equal quantities selected at random throughout the lot in the frequency listed below. The composite

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shall be placed in a clean, dry glass container, sealed and marked for testing. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if the composite fails to meet any test requirement specified.

Lot size (gallons)	Number of units to form composite
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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INSTRUCTIONS FOR TESTING MATERIALS AND COMPONENTS

TABLE V

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations per Comp. Samp.	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Composition, percent by weight:									
2-Butyl-2-ethyl-1, 3-propanediol	Table I	4.1.2		X	2		0.5 percent		
Benzyl benzoate	Table I	4.1.2		X	2		0.5 percent		
N-Butyl acetanilide	Table I	4.1.2		X	2		0.5 percent		
Polyoxyethylene (20) sorbitan monooleate	Table I	4.1.2		X	2		0.1 percent		
2-Butyl-2-ethyl-1, 3-propanediol:									
Color	Table II	4.4.1.3		X	2	X			
Freezing point	Table II	4.4.2.2		X	2		0.1°C.		
Free acidity as propionic acid	Table II	4.4.7		X	2		0.01 percent		
Benzyl benzoate:									
Color	Table III	4.4.1.3		X	2	X			
Setting point	Table III	4.4.2.1		X	1		0.1°C.		
Specific gravity 25°/4°C.	Table III	4.4.3		X	2		0.001		
Refractive index n/D 20°C.	Table III	4.4.4		X	2		0.001		
Total chlorine	Table III	4.4.5.1		X	2		0.01 percent		

AMXRE Form 598
8 February 1965

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INSTRUCTIONS FOR TESTING MATERIALS AND COMPONENTS

TABLE V (cont'd)

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations per Comp. Samp.	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Benzyl benzoate: (cont'd)									
Side chain chlorine	Table III	4.4.5.2		X	2		0.01 percent		
Ester content	Table III	4.4.6		X	2		0.1 percent		
Free acidity as benzoic acid	Table III	4.4.7		X	2		0.01 percent		
N-Butyl acetanilide:									
Color	Table IV	4.4.1.3		X	2	X			
Setting point	Table IV	4.4.2.3		X	1		0.1°C.		
Boiling range:									
Up to 278°C.	Table IV	4.4.8		X	1		0.1 percent		
Up to 283°C.	Table IV	4.4.8		X	1		0.1 percent		
Free acidity as acetic acid	Table IV	4.4.7		X	2		0.01 percent		
Polyoxyethylene (20) sorbitan monooleate:	3.1.4	1/							
Water content	3.1.4	4.4.12		X	2		0.01 percent		

1/ Supplier shall furnish a certificate of compliance covering this requirement.

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4.3 Inspection of the end item.-

4.3.1 Examination of the end item.- The end item shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.4. The lot size shall be expressed in units of filled unit containers of the specified capacity for the examinations in 4.3.1.1 and 4.3.1.2 and in units of shipping containers for the examination in 4.3.1.3.

4.3.1.1 Examination of the end item for visual defects.- The sample unit for this examination shall be one filled unit container.

Examine	Defect
Container	Not size, shape, type container specified. Exterior coating not as specified (see 5.1.1). Any evidence of seepage or leakage from any part of container including closure and seams. Dented and distorted container.
Workmanship	Contents not homogeneous. Presence of foreign matter. Not clear.
Markings	Missing, incomplete, incorrect, not as specified (see 5.3.1). Not affixed in place and coated with water-resistant adhesive as specified.

4.3.1.2 Examination for average net contents.- The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net contents per container for all sample units examined is less than specified or indicated.

4.3.1.3 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing and marking requirements comply with the requirements of Section 5 of this specification. The sample unit shall be one shipping container, fully prepared for delivery except that it need not be sealed. The lot shall be the number of shipping containers in the end item inspection lot. Closed shipping containers shall be examined for closure defects.

Examine	Defect
Markings	Incorrect; incomplete; illegible; omitted; of improper size, sequence or method of application; not in accordance with contract requirements.

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Examine	Defect
Material	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components such as: incomplete closure of case liners, container flaps, loose or inadequate sealing, strapping or stapling. Bulged or distorted container.
Contents	Number of primary containers per shipping container is less than specified.

4.3.1.4 Inspection levels and acceptable quality levels (AQLs) for examination.- The inspection levels for determining the sample size and acceptable quality levels (AQLs), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1.1	I	2.5
4.3.1.2	S-2	N.A.
4.3.1.3	S-2	2.5

4.3.2 Testing of the end item.- The end item shall be tested for the characteristics specified in table VI. The lot size shall be expressed in gallons. The sample size for testing shall be a pint composite. The composite shall be made by thoroughly mixing equal quantities selected at random throughout the lot in the frequency listed below. The composite shall be placed in a clean, dry glass container, sealed and marked for testing. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if the composite fails to meet any test requirement specified.

Lot size (gallons)	Number of units to form composite
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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INSTRUCTIONS FOR TESTING OF THE END ITEM

TABLE VI

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations		Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver	per	Comp. Samp.	Pass or Fail	Numerically to Nearest		
Homogeneity	3.3	4.4.9			1		X			
Emulsibility	3.4	4.4.11			1		X			

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8 February 1965

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4.4 Test procedures.-

4.4.1 Color.-

4.4.1.1 Preparation of Platinum Cobalt Standard (Hazen).- Dissolve 1.245 g potassium chloroplatinate (K_2PtCl_6) containing 0.5 g platinum and 1 g of crystallized cobaltous chloride ($CoCl_2.6H_2O$ containing approximately 0.248 g of cobalt) in 100 ml concentrated hydrochloric acid and dilute with distilled water to make one liter. This solution has a Hazen color of 500.

4.4.1.2 Preparation of comparison standards for components benzyl benzoate, 2-butyl-2-ethyl-1, 3-propanediol, and N-butyl acetanilide.- Standards having Hazen color numbers 50, 100, 150, 200, 250, 300 are made by diluting in standard tall form 50 ml Nessler tubes 5, 10, 15, 20, 25 and 30 ml of the standard solution (see 4.4.1.1) with distilled water to make 50 ml.

4.4.1.3 Benzyl benzoate, 2-butyl-2-ethyl-1, 3-propanediol, and N-butyl acetanilide.- Place 50 ml of benzyl benzoate, 2-butyl-2-ethyl-1, 3-propanediol, and N-butyl acetanilide in a matched Nessler color comparison tube and match with comparison standards (see 4.4.1.2). Use a colorimeter provided with white light.

4.4.2 Setting point and freezing point.-

4.4.2.1 Benzyl benzoate.- Fill a 1 by 8 inch test tube with sample to a depth of about 3-1/2 inches and cool in a water bath to about 18°C. Place the tube containing the cooled sample inside an air jacket (a 2-inch by 7-inch test tube) supported in a water bath maintained at 13° ± 2°C. Seed the sample with a small amount of material from a previously frozen sample and stir using a glass ring stirrer and a thermometer graduated in fifths of a degree, until the temperature of the sample rises to a maximum. This will require a considerable period of time, sometimes as much as 15 to 30 minutes, because of the very slow rate of crystallization and low latent heat of fusion of benzyl benzoate. Record the maximum temperature reached as the setting point and report to the nearest 0.1°C.

4.4.2.2 2-butyl-2-ethyl-1, 3-propanediol.- The 2-butyl-2-ethyl-1, 3-propanediol shall be tested as specified in 4.4.3.1 except the sample shall be cooled to 43°C. and the water bath shall be maintained at 35°C.

4.4.2.3 N-butyl acetanilide.- The N-butyl acetanilide shall be tested as specified in 4.4.2.1 using a seed crystal prepared by freezing a sample cooled in a dry ice acetone bath.

4.4.3 Specific gravity.- The specific gravity shall be determined by any method accurate to the third decimal point.

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4.4.4 Refractive index.- The refractive index shall be determined by any method accurate to the fourth decimal point.

4.4.5 Chlorine content.-

4.4.5.1 Total organic bound chlorine.- Transfer a quantity containing 1 g accurately weighed, of the material to a 250 to 500 ml standard tapered Erlenmeyer flask. Add 25 ml of 99 percent isopropanol. Add 2.5 g of metallic sodium cut in small pieces, and shake the flask to mix the sample with the alcohol. Connect to a reflux condenser and boil gently for 1/2 hour. Shake the flask occasionally. Eliminate all excess metallic sodium by cautiously adding about 10 ml of 50 percent isopropanol through the condenser at the rate of 1 to 2 drops per second. Boil for an additional 10 minutes; then add 60 ml of distilled water. Transfer to a 400 ml beaker and heat on a steam bath for 30 minutes, cool, add 2-3 drops of phenolphthalein solution, neutralize by adding nitric acid (1-1), and then add 10 ml in excess. Add 5 ml of 0.01 N silver nitrate solution and coagulate the precipitated silver chloride by digesting in a steam bath for 1/2 hour with frequent stirring. Cool, filter through a No. 42 Whatman filter paper or equivalent, and wash thoroughly with distilled water. Add 5 ml of saturated ferric alum solution and determine excess silver nitrate in the filtrate by titration with 0.02 N potassium thiocyanate. Compute the net number of ml of 0.01 N silver nitrate consumed by the sample. Calculate percent chlorine as follows:

4.4.5.1.1 Calculation.-

$$\text{Percent chlorine} = \frac{\text{Milliliter 0.01 N } AgNO_3 \text{ used X } 0.0354}{\text{Weight of sample (in grams)}}$$

Note: A blank determination (without sample) should be made following the exact procedure given above but limiting the 0.01 N silver nitrate solution to 5 ml in order to obtain a chloride correction value for all reagents used.

4.4.5.2 Side chain chlorine.- Place a solution of 15.00 g of benzyl benzoate in 50 ml of methanol in a 250 ml standard taper Erlenmeyer flask. Add 60 ml of methanolic 1.5 N sodium methylate. Add boiling chips to the flask and then connect the flask to a water cooled standard taper reflux condenser and boil the flask contents for one hour. Cool the flask and wash the condenser down with 25 ml distelled water. Remove the condenser from the flask, add 2 drops of phenolphthalein indicator solution, made acid with 3 N nitric acid. Add an excess of 20 ml nitric acid. Then add 10 ml of 0.05 N silver nitrate. Let stand for 5 minutes and then add 2 ml of nitro-benzene and 0.5 ml of 10 percent ferric nitrate solution. Titrate the excess silver nitrate with 0.05 N potassium thiocyanate. A blank determination should be made following the exact procedure given above to obtain a chloride correction value for all reagents used. Calculate the net number of ml of 0.05 N silver nitrate consumed by the

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sample and by the blank determination. Calculate the g of chlorine in the sample and in the blank, subtract the blank value from the sample value, and calculate the percent chlorine in the sample.

One ml of 0.05 N $AgNO_3$ = 0.001773 g chlorine

Note: The sodium methylate solution is made by dissolving 34.5 g of sodium in anhydrous methanol and diluting to a 1 liter.

4.4.6 Ester content.- The ester content shall be determined in accordance with Method 5171 of Fed. Test Method Std. No. 141.

4.4.7 Free acidity.- The free acidity shall be determined in accordance with Method 5252 of Fed. Test Method Std. No. 141. The milliequivalent value (E) for propionic acid is 0.074 g, for benzoic acid 0.123 g, for acetic acid 0.060 g.

4.4.8 Distillation.- The distillation shall be determined in accordance with ASTM Designation D-86-62 with correction applied to the distillation thermometer to correct for emergent stem as follows:

$t_c = 0.000156\ L\ (t_o - t_e) + t_o$.

t_c = corrected temperature.

L = length of emergent stem measured in scale degrees.

t_o = observed temperature

t_e = average temperature of emergent, mercury column.

4.4.9 Homogeneity.- Homogeneity shall be determined by visual inspection. After standing at room temperature for 1 hour, there shall be no separation of liquid or solid phases.

4.4.10 Hard water.- The hard water specified in 4.4.11 shall have the following composition, and is designed to provide a hardness of 342 parts per million (ppm) calculated as calcium carbonate:

$CaCl_2$ (anhydrous)	0.3037 g
$MgCl_2.6H_2O$	0.1388 g
Distilled water to make	1 liter

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4.4.11 Emulsibility.- Add 75 ml of hard water (see 4.4.10) add 15 ml of the sample into a 100 ml glass-stoppered graduated cylinder. An emulsion shall be formed by shaking the cylinder in an arc 1 foot in extent for a total of 30 shakes at the rate of three shakes per second. Any creaming which may form after standing for 30 minutes shall readily re-emulsify when the cylinder is given three shakes of the above specified type. There shall be no separation of phase during an 8-hour period of standing. Any creamy layer or layers formed during the 8-hour period shall readily re-emulsify when given three shakes of the above specified type.

4.4.12 Emulsifier water content.- The water content shall be determined by titration with Karl Fischer reagent in accordance with Method 4082 of Fed. Test Method Std. No. 141.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- One gallon of repellent shall be filled into a round, square, or rectangular base one-gallon capacity metal can conforming to type I of PPP-C-96, except that black plate or 0.25-pound electrolytic tin plate shall not be used. Exterior protective coating shall conform to Plan B.

5.1.2 Level C.- One gallon of repellent shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.- Six gallons of repellent packaged as specified in 5.1, shall be packed in accordance with level A packing requirements in appendix of PPP-C-96. Except when fiberboard boxes conforming to PPP-B-636 are used, they shall be made from materials conforming to grade V2s and shall be fitted with a one-piece taped corrugated liner and top and bottom pads made from material conforming to grade V3c of PPP-F-320. Top and bottom pads shall be cut full inside length and width of the fiberboard container. The height of the liner shall be equal to the inside depth of the container less the thickness of the top and bottom pads.

5.2.2 Level B.- Six gallons of repellent, packaged as specified in 5.1, shall be packed in accordance with the level B packing requirements in the appendix of PPP-C-96.

5.2.3 Level C.- The repellent, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

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5.3 Marking.- In addition to the labeling specified in 5.3.1 and any special marking required by the contract or order, each unit container and shipping container shall be marked in accordance with MIL-STD-129, including the batch number.

5.3.1 Labeling.- The label data, as specified in 3.4, shall be durably and legibly marked on the main panel of the body by lithographing, stenciling, or printing with a black, noncorrosive, weather-resistant, waterproof ink as specified in MIL-STD-129. If all of the label data cannot be applied due to being too voluminous, that portion such as the directions for use, may be placed on a label, the label securely affixed in place, and waterproofed as specified in MIL-STD-129.

6. NOTES

6.1 Intended use.- Insect repellent specified in this specification is intended to be used for impregnation of clothing to protect against mites, fleas, ticks, mosquitoes, and land leaches (see 3.4).

6.2 Ordering data.- Purchasers should exercise any desired operations offered herein and procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Custodians:

Army - GL

Review activities:

Army - GL, MD, MU

User activities:

Army - ME
Navy - MC, MS, SH, YD

Preparing activity:

Army - GL

Project No. 6840-0178

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Insect Repellent, Clothing Application (Formula M-1960) - MIL-I-12123D

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01760

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DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-QS
Natick, Massachusetts 01760

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